BR

Bulk Repair

INVENTORY PORTAL

Complete User Guide

Har tab, har screen, har field — step-by-step Hinglish guide. Inventory, imports, rules, reports, purchasing, security aur troubleshooting.

Edition	Version	Date	Applies to
Hinglish	1.0.1	19 October 2026	Portal · Desktop · Android

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Bulk Repair — Complete User Guide (Hinglish)

Version 1.0.1 · Portal + Desktop (Windows/macOS) + Android

Ye guide app ke **har tab, har button aur har field** ko explain karta hai — kaam kya hai, kaun use karega, step-by-step kya karna hai, aur kis condition me kya hoga.

> **Legend:** ✅ Local mode + Apps Script mode dono me available · ⚠️ Sirf Apps Script (Google Sheets) mode me · 🖥️ Sirf desktop app me

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1. Ye guide kaise use karein

AAP KYA KAR RAHE HO	KAHAN PADHO
Pehli baar install/setup	Section 3 → 4 → 5
Roz ka kaam (stock, import, labels)	6.2, 6.3, 6.6, 6.13
Reports/management	6.8 – 6.12, 6.15
Admin kaam (users, security)	6.17 – 6.26
Barcode scanner lagana	Section 7
Kuch kaam nahi kar raha	Section 10 aur 11
Naya kya hai 1.0.1 me	Section 15

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2. App kya hai — modes aur components

Bulk Repair ek **warehouse / repair inventory system** hai:

- Devices ka master data (barcode, model, brand, status, age, location, priority)
- Excel/CSV/PDF se bulk import + rollback
- Rules engine se auto-classification (age, status, location)
- Dashboard, reports, labels, alerts
- Vendors aur purchase orders (receiving se stock update)
- Users, roles, sessions, audit, 2FA

2.1 Do data modes

	LOCAL	APPS SCRIPT
Data kahan	`bulkrepair.db` (SQLite)	Google Sheet
Internet	Nahi chahiye	Chahiye
Speed	Fast	Network pe depend
Multisource merge	✅ file-based	✅ sheet-based
Google Sheet se direct import	❌	✅
Sheet security (lock/unlock)	❌	✅
Baaki sab tabs	✅	✅

2.2 Components

COMPONENT	KYA HAI	DATA KAHAN
Portal (web)	Browser me poora UI	Backend se
Desktop app	Windows/macOS app; **server khud start karta hai**	App data folder
Android APK	Mobile UI + camera scan	External backend
Server	Node.js + Express + SQLite	`data/bulkrepair.db`
GAS backend	Apps Script + Sheets	Google Sheet

2.3 Desktop data location

OS	DATABASE	BACKUPS	LOGS
macOS	`~/Library/Application Support/Bulk Repair/data/bulkrepair.db`	`.../data/backups/`	`.../logs/server.log`
Windows	`%APPDATA%\Bulk Repair\data\bulkrepair.db`	`...\data\backups\`	`...\logs\server.log`

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3. Install aur first run

3.1 Desktop app (sabse easy)

1. `dist-builds/` se apna installer lo:
   - Windows x64: `BulkRepair-1.0.1-windows-x64-portable.exe`
   - Windows ARM: `BulkRepair-1.0.1-windows-arm64-portable.exe`
   - macOS: `BulkRepair-1.0.1-macos-arm64.zip`

2. **Windows:** exe double-click → SmartScreen aaye to **More info** → **Run anyway**
3. **macOS:** zip unzip → `Bulk Repair.app` ko Applications me drag → pehli baar **right-click** → **Open**
4. App khulte hi andar server start hota hai (port 5050, busy ho to 5051…)
5. Setup wizard → owner account → Dashboard

> Desktop me alag se Node/npm/server install karne ki zarurat **nahi**.

3.2 Server + portal (browser)

```
cd inventory-automation/server
npm install
npm run dev
```

- URL: `http://localhost:5050`
- Port `.env` se aata hai
- Pehli baar DB khud banti hai

Production:

```
npm run build:panel   # portal build + copy
npm run build
npm start
```

3.3 Android APK

```
adb install -r BulkRepair-1.0.1-android-debug.apk
```

- USB: `adb reverse tcp:5050 tcp:5050` → app me `http://localhost:5050`
- Wi-Fi: `http://<computer-ka-LAN-IP>:5050`
- Backend kisi aur machine/VPS pe ho to wahi URL do

APK ke saath **camera scan** chalta hai; desktop pe hardware scanner.

3.4 Docker

```
cd inventory-automation
docker compose up --build
```

- URL: `http://localhost:5000`
- Data `/data` volume me rehta hai

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4. First-time setup — step by step

Step 1 — Backend choose karo

OPTION	KAB CHUNO
This computer (offline)	Data local chahiye, internet nahi
Google Apps Script / Google Sheets	Data Sheet me, phone se kahin se access

- Local choose kiya: **Server URL blank** chhod do (desktop ka embedded server khud use hoga)
- External backend: `http://192.168.1.10:5050` ya `https://your-domain.com`
- GAS choose kiya: deployment ka `/exec` URL daalo → **Test connection**

Test connection ke results:

RESULT	MATLAB
Connected ✅	URL sahi, backend chal raha hai
Not reachable ❌	URL galat, server band, ya firewall

Step 2 — Google Sheet (optional)

Sirf quick-access link save hota hai. Local mode me skip.

Step 3 — Owner account

FIELD	RULE
Email	Valid email
Password	**10+ characters**, letters + numbers dono
Confirm	Same password

Finish setup → pehla account `OWNER` role ke saath banta hai.

Conditions:

SITUATION	KYA HOGA
Pehla account	Owner create hoga
Pehle se users hain	Bootstrap fail → wahi email/password se **login fallback**
Password weak	Inline error, aage nahi badhega
Email invalid	Inline error

5. Login, roles aur navigation

5.1 Login

- Email + password
- 2FA on hai to 6-digit code (ya recovery code)
- **Keep me signed in**
- 5 galat attempts → **15 minute lock**
- Password bhool gaye → admin se reset karwao

5.2 Roles

ROLE	KYA KAR SAKTA HAI
VIEWER	Sab dekh sakta hai, kuch badal nahi sakta
OPERATOR	Inventory edit, imports, labels, scanner, AI commands
MANAGER	+ reports, schedules, purchasing, activity/history
ADMIN	+ users, security, settings, email config, system health
OWNER	Sab kuch

5.3 Sidebar — 28 tabs

GROUP	TABS
Overview	Dashboard
Operations	Data Sources, Import Data, Multisource, Field Mapping, Inventory, Rules, Reports, Report Templates, Report Builder, Alerts, Ageing Trends, Labels, Schedules, Exports, Purchasing
Administration	Users & Roles, Security, Sessions & Audit, Activity Log, Processing, Processing History, Data Quality, System Health, Email Settings, Settings
Tools	AI Command Center, Help

Sidebar ke bottom me **language** (English / हिन्दी / ગુજરાતી) aur **Collapse** button hai.

5.4 Collapse behaviour

- Collapse karne pe sidebar **76px** ho jaata hai
- Sirf icons dikhte hain, hover pe tooltip
- Language selector hide ho jaata hai
- Expand pe sab wapas

5.5 Common login conditions

CONDITION	RESULT
Account inactive	Login block; purane sessions bhi revoke
Session revoke hua	Agli request pe login page
Role change hua	Agli request se naya role
Backend band	"Not connected" error
2FA code galat	`Invalid authenticator code` + attempt count

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6. Har tab ka detailed guide

Har tab ke liye: **kaam**, **kaun**, **page pe kya hai**, **steps**, **conditions**, **galtiyan**.

6.1 Dashboard

Kaam: ek nazar me poora inventory status.

Kaun: sab roles.

Page pe kya hai:

BLOCK	KYA DIKHATA HAI
4 KPI cards	Inventory items, Active sources, Active reports, Recent errors
By status	Status distribution (bar)
By age bucket	Age buckets (donut)
By status bucket	Bucket distribution (bar)
Top brands	Top 5 brands (bar)
Ageing trend	Daily line chart
Recent runs	Latest import/process jobs
Recent errors	Last errors (barcode, field, message)

Steps:

1. Page kholo — data khud load hota hai
2. **Refresh** — latest data
3. **Run daily process** (OPERATOR+) — rules dobara apply + trend snapshot

Conditions:

CONDITION	RESULT
Inventory empty	Charts "No data yet"
Daily process nahi chala	Trend empty
Errors hain	Recent errors count red
Backend down	Error panel + Retry

Tips: Subah sabse pehle errors aur ageing dekho. Charts empty ho to pehle import karo, phir process chalao.

6.2 Data Sources

Kaam: batana ki data kahan se aayega.

Kaun: ADMIN (view sab kar sakte hain).

Page pe kya hai: sources ka table (name, role, type, sheet, join key, active) + New source.

Steps:

1. **New source**
2. Name: `Daily QC`
3. Role: `PRIMARY` (main data) ya `JOIN` (extra data)
4. Type: free text (`Google Sheet` , `QC` , `TRC` , `Inventory` , `Location`)
5. Sheet name: `Sheet1`
6. Sheet ID: external sheet ho to
7. Join key: `barcode`
8. Join fields: `STOCK_AGE,LOCATION`
9. Active ON → **Save**

Conditions:

CONDITION	RESULT
Name duplicate	Save fail (`already exists`)
Active OFF	Pipeline us source ko skip karega
Delete	Mapping bhi delete; inventory safe
Vendor/PO wala source?	Koi lena-dena nahi

Tips: Type me `QC` likhne se system source type `QC_STOCK` infer karta hai — merge/priority ke liye zaroori.

6.3 Import Data (Universal import) ✅

Kaam: file se bulk data laana.

Kaun: OPERATOR+ (view sab).

Supported files: CSV, TSV, TXT, XLS, XLSX, PDF · max **15 MB / 50,000 rows**.

Step-by-step:

1. **Import Data** tab
2. **Choose file**
3. Columns auto-detect hote hain — mapping check karo
4. `BARCODE` required hai
5. **Validate** — errors preview
6. **Import now**
7. Result card: processed / valid / imported / failed / skipped
8. Galti ho to **Roll back import**

Conditions:

CONDITION	RESULT
Barcode column missing	Validate fail
Duplicate barcode	Existing row **update** hota hai
Empty incoming value	Purani value preserve hoti hai
File bada	Upload reject
Partial errors	Job `PARTIAL_FAILURE`, baaki rows import
Rollback	Naye barcodes delete, purane restore
Dobara rollback	`already rolled back` error

Tips: Pehle 10–20 rows ki test file se mapping verify karo. Age numeric rakho.

6.4 Multisource ✅

Kaam: QC + TRC + Inventory + Location ko **barcode pe merge** karna.

Kaun: OPERATOR+.

Local mode:

1. Har source ke liye ek file choose karo
2. **Validate files** — stage + auto mapping
3. **Import all sources**
4. Result: devices, inserted, updated, conflicts
5. **Roll back import** available

Apps Script mode: sheet names/IDs do → validate → import.

Conditions:

CONDITION	RESULT
Same barcode, same field, different value	Conflict record
Conflict winner	Priority rules decide
Barcode missing	Row reject
Merge ke baad	Inventory upsert, movement history banti hai

6.5 Field Mapping

Kaam: source column ko system field se jodna.

Kaun: ADMIN.

Steps:

1. Source choose karo
2. Har row: source column → target field
3. Required checkbox
4. Output order
5. Active
6. **Save mapping**

Common target fields: `BARCODE`, `MODEL`, `BRAND`, `STATUS`, `STOCK_AGE`, `LOCATION`, `PRIORITY`, `BIN_BUCKET`.

Conditions:

CONDITION	RESULT
`BARCODE` mapped nahi	Import fail
Ek column do targets	Last wala jeetega — avoid karo
Empty target	Column ignore
Mapping save	Agla import naye mapping se chalega

6.6 Inventory

Kaam: master data dekhna aur edit karna.

Kaun: view sab; edit OPERATOR+.

Toolbar: Search, Bulk edit, Scan/Scanner ready, filters (Status, Age, Brand, Category, Priority, Bin, Source), Clear, Views, Columns, Export, Add row.

Saved views:

1. Filters set karo
2. **Views → Save current view…**
3. Naam do → Save
4. Dobara use: Views → naam

5. Delete: X

Search/filters/page size **automatic save** hote hain.

Columns: hide/show; `BARCODE` lock.

Row drawer: saare fields + **History** timeline (Added/Removed/field old→new) + Edit/Delete.

Add/Edit conditions:

FIELD	RULE
BARCODE	Required, unique, edit me change nahi
STOCK_AGE	Days (number)
AGE_BUCKET / PRIORITY / STATUS_BUCKET	Rules se auto; blank chhod do
LOCATION / BIN_BUCKET	Rack/shelf

Bulk edit: rows select → status/location/bin/priority → apply. Sirf selected fields badalte hain; audit log entry banti hai.

Delete: permanent; history me `Removed` record.

Desktop scanner: page pe kahin bhi scan karo → search fill + toast.

6.7 Rules

Kaam: age/status/location se auto buckets aur priority.

Kaun: ADMIN.

Teen tabs:

TAB	FIELDS	EXAMPLE
Age buckets	Min age, Max age, Bucket, Priority	0–29 → Below 30 → P3
Status buckets	Raw status, Bucket, Marking	`QC Fail` → `QC_FAIL`
Location rules	Pattern, Bin bucket, Condition	`SW*` → Shelf A

Steps:

1. Tab kholo
2. **Add row**
3. Values bharo
4. **Save**
5. **Processing → Run daily process**

Conditions:

CONDITION	RESULT
Rule overlap	Pehla match jeetta hai
Save ke baad	Agle process run pe apply
Match nahi mila	Field khali rehta hai
Priority rule (GAS)	Merge winner decide karta hai

Tips: Age ranges continuous rakho (0–29, 30–59, 60–74, 75+).

6.8 Reports

Kaam: grouped counts aur exports.

Kaun: MANAGER+ (VIEWER dekh sakta hai).

Group by: status, brand, age_bucket, status_bucket, location, bin_bucket, model.

Filters: status, brand, age bucket, status bucket, location.

Steps:

1. Group by choose karo
2. Filters lagao
3. **Run**
4. Export: **CSV**, **Excel**, **PDF**

Conditions:

CONDITION	RESULT
Data empty	0 rows report
Filter galat	Empty result
Export	File download
Export all to sheets ⚠️	Sirf GAS

6.9 Report Templates

Kaam: reusable report formats save karna.

Kaun: ADMIN (view MANAGER+).

Page: Reports page ka focused templates mode.

Fields:

FIELD	MATLAB
Name	Template ka naam
Config JSON	Group by + filters JSON
Active	ON/OFF

Steps:

1. **New template**
2. Name + Config JSON
3. Save
4. Edit/Delete row actions

Conditions: Invalid JSON → save reject. Local mode supported.

6.10 Report Builder

Kaam: naya report definition banana (saved reports list ke saath).

Kaun: MANAGER+.

Steps:

1. **New report**
2. Naam
3. Filter field (`BRAND`) + value (`Apple`)
4. Report columns (`STATUS,AGE_BUCKET`)
5. Save
6. Row pe **Generate** / Edit / Delete

Conditions: naam duplicate → update ho jaata hai. Empty columns → default.

6.11 Alerts

Kaam: threshold alerts banana aur test karna.

Kaun: ADMIN (view sab).

Alert types:

TYPE	MATLAB
`AGE_OVER`	Itne din se purana stock
`STATUS_BUCKET_COUNT`	Bucket count threshold
`MISSING_FIELD`	Field missing

Fields: Name, Filter field, Filter value, Operator, Threshold, Email to, Telegram, Active.

Steps:

1. **New alert**
2. Values bharo (`e.g. 50`)
3. Save
4. **Check now** — triggered list
5. Email/WhatsApp test buttons

Conditions:

CONDITION	RESULT
SMTP missing	Test fail with clear message
Threshold cross	Alert triggered + `last_fired_at` update
Active OFF	Skip

6.12 Ageing Trends

Kaam: daily ageing trend aur current buckets.

Kaun: sab.

Page pe: line chart (date + buckets) + current age bucket table.

Conditions: Daily process nahi chala → "No trend data yet". Inventory empty → table empty.

6.13 Labels

Kaam: barcode labels print karna.

Kaun: OPERATOR+.

Settings: count (1–200), size (50×25 / 38×25 / 60×30 mm), columns (2/3/4), barcode height, fields (barcode text, brand+model, location+bin).

Steps:

1. Settings set karo
2. **Print**
3. Print dialog: margins **None**, scale **100%**, background ON

Conditions: Barcode missing → blank label. Cut ho raha hai → height kam / size bada.

6.14 Schedules

Kaam: recurring PDF/XLSX reports email karna.

Kaun: MANAGER+ view; ADMIN create.

Fields: Name, Group by, Frequency (Daily/Weekly/Monthly), Format (PDF/XLSX), Recipients, filters, Active.

Steps:

1. **New schedule**
2. Values bharo

3. Save
4. **Run** — turant send
5. Edit/Delete

Conditions:

CONDITION	RESULT
SMTP missing	Run fail (clear error)
Due schedule	Server har 30 min check karta hai
Restart	`last_run_at` se duplicate nahi
Multiple recipients	Comma-separated

6.15 Exports

Kaam: grouped reports ya poora inventory download karna.

Kaun: sab.

Page pe: group-by select + **CSV / Excel / PDF** buttons + **Inventory CSV**.

Conditions: Empty group → empty file. Backend auth fail → error toast.

6.16 Purchasing

Kaun: MANAGER+.

Vendors tab: name, contact, email, phone, address, notes, active. Delete tabhi jab PO na ho.

Purchase orders tab: PO number, vendor, status, lines, ordered, received, expected.

Steps (PO):

1. **New purchase order**
2. Vendor + status (Draft/Ordered) + expected date + notes
3. Lines: barcode (optional), model, brand, qty, unit cost
4. Save → PO number auto `PO-YYYYMMDD-001`
5. **Receive** → quantities → **Receive stock**

Conditions:

CONDITION	RESULT
Over-receive	Reject with remaining qty
Barcode wali line	Inventory me `IN_STOCK`
Sab receive	Status `RECEIVED`
Partial	`PARTIAL`
Received order edit/cancel	Block
Vendor delete with PO	Block

6.17 Users & Roles

Kaun: ADMIN.

Table: email, role, active, password set?, locked until, last login.

New user: email + role + password (10+ letters/numbers).

Edit: role change; active OFF → login block + sessions revoke; apna role change nahi.

Reset password: naya password; us user ke sab sessions revoke.

Conditions:

CONDITION	RESULT
Weak password	Reject
Email duplicate	Update existing
Active OFF	Login block
Locked account	15 min wait / admin unlock

6.18 Security

Kaam: 2FA status, password policy, active sessions.

Kaun: ADMIN.

Page pe:

CARD	KYA
Two-factor authentication	Enabled/off + recovery codes left + Settings link
Password policy	10+ chars, letters+numbers, lockout rule
Active sessions	user, role, signed in, last seen, **Sign out**

Conditions: 2FA reset admin se; session revoke → token dead.

6.19 Sessions & Audit

Kaun: ADMIN.

Active sessions: per-session sign out, per-user **All**.

Login history: last 100 events (OK / FAILED / LOCKED / BLOCKED / LOGOUT) + IP.

Kab dekhna: roz shaam, employee exit, suspicious logins.

6.20 Activity Log

Kaam: har action ka audit trail.

Kaun: MANAGER+.

Page: activity table (time, actor, action, entity, details).

Conditions: Actions: login, inventory changes, imports, PO, settings, user changes.

6.21 Processing

Kaam: daily rules pipeline trigger karna.

Kaun: OPERATOR+.

Steps: Run daily process → confirm → rules apply + trend snapshot + alerts check.

Conditions: Data empty → process chalega par counts 0. Error → toast.

6.22 Processing History

Kaam: import/process jobs ka record.

Kaun: MANAGER+.

Page: runs table (file, kind, status, rows, created).

Conditions: Status: COMPLETE / PARTIAL_FAILURE / UPLOADED / DISCARDED.

6.23 Data Quality

Kaam: missing fields / errors ka overview.

Kaun: MANAGER+.

Page: quality summary + errors by source.

Conditions: Score missing fields ke ratio se. Naya import → refresh.

6.24 System Health

Kaam: backend status + self-test.

Kaun: ADMIN.

Page: health JSON + checks table:

CHECK	KYA
Database connection	SQLite online
User accounts	count
Inventory table	count
Email configuration	SMTP host set?
Automatic backups	every 6h, 7 kept

Conditions: FAIL check → detail me reason.

6.25 Email Settings

Kaam: SMTP configure karna (alerts + scheduled reports).

Kaun: ADMIN.

Fields: SMTP host, port, user, password, from, alert recipients.

Steps:

1. Values bharo (`smtp.gmail.com`, `587`, app password)
2. **Save**
3. **Send test**

Conditions:

CONDITION	RESULT
Host/user galat	Test fail with error
Gmail	App password chahiye
Save	Turant apply

6.26 Settings

Kaun: ADMIN.

Sections:

SECTION	KEYS
System settings	Telegram, backup, SMTP, WhatsApp, Google, LLM
Notification tests	Email / WhatsApp
Connection	API URL, Sheet URL
Two-factor authentication	Enable/disable + recovery codes
Barcode scanner	Mode, min length, terminator, test, serial connect
Change password	Current + new (10+ letters/numbers)

Conditions: Secret fields masked; save ke baad test karo.

6.27 AI Command Center

Kaun: OPERATOR+.

Commands (English/Hindi):

- `List Apple devices older than 75 days`
- `How many devices are in Bin Storage?`
- `Show today's inventory summary`

Kaise chalta hai: command → filters parse → inventory query → result.

Conditions:

CONDITION	RESULT
LLM configured	Smarter parsing
LLM fail	Rules parser fallback
Kuch samajh nahi aaya	`success: false` message

6.28 Help

Kaun: sab.

Page pe: getting started, barcode scanners, backups, troubleshooting, documentation link.

7. Hardware barcode scanners (desktop)

Desktop (Windows/macOS) pe camera scan nahi — **hardware scanner** support hai.

7.1 USB HID scanner (99% scanners)

- Scanner **keyboard ki tarah** kaam karta hai — **driver install nahi karna**
- App fast keystrokes detect karta hai (average ≤45ms per character)

- Enter/Tab terminator pe search fill hota hai
- Inventory page pe **kahin bhi** scan karo
- Toolbar me **Scanner ready** badge dikhta hai

7.2 Serial / COM scanner

1. Scanner ko serial mode me set karo (vendor manual)
2. **Settings → Barcode scanner → Connect serial scanner**
3. Port select karo (default 9600 baud)
4. Scan karke test karo

7.3 Scanner settings

SETTING	MATLAB	DEFAULT
Mode	Hardware / Camera	Hardware
Minimum code length	Chhoti typing ignore	6
Terminator	Enter / Tab / None	Enter
Test scanner	Live test field	—

7.4 Conditions

CONDITION	RESULT
Normal typing	Ignore (speed check)
Modal/drawer open	Scan ignore
Code < min length	Ignore
Scanner sends no terminator	Detect nahi hoga — Terminator setting badlo
Mobile app	Camera scan available

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8. Daily / weekly workflow

Subah

1. **Dashboard** — errors, ageing, counts
2. **Sessions & Audit** — failed logins (admin)

Naya stock

1. **Import Data** → file → Validate → Import
2. Errors check karo
3. **Labels** print

4. **Processing** → **Run daily process** (buckets refresh)

Device movement

1. **Inventory** → scan/search
2. Row → Edit → location/status
3. History me change confirm

Bulk update

1. Rows select → **Bulk edit**
2. Fields choose → apply

Order

1. **Purchasing** → New PO → Save
2. Receive hone pe **Receive** → inventory update

Shaam / weekly

1. **Reports** → PDF/XLSX
2. **Schedules** me weekly report
3. **Data Quality** + **Processing History** review
4. **Sessions & Audit** check
5. Backup verify

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9. Backup, restore aur rollback

9.1 Automatic backup

- Server boot pe + har **6 ghante**
- Latest **7** copies
- Location: DB folder ke andar `data/backups/`

9.2 Manual backup

- Desktop: `data` folder copy karo
- Server: `server/data/bulkrepair.db` copy karo

9.3 Restore

1. Server/app band karo
2. `bulkrepair.db` rename karo
3. Backup file ko `bulkrepair.db` naam se copy karo
4. Start karo

9.4 Import rollback

- Sirf **committed** job ka
- Naye barcodes delete, purane restore
- Dobara rollback nahi
- Dry run ka rollback nahi

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10. Condition cookbook — "agar aisa ho to kya hoga"

CONDITION	RESULT
Duplicate barcode import	Existing row update
Barcode missing	Row reject
Age text me	Rule match nahi, bucket khali
50k+ rows	Upload reject
Import ke baad galti	Rollback (ek baar)
Rollback dobara	`already rolled back`
Vendor ke paas PO	Delete block
PO se zyada receive	Reject + remaining
Received PO cancel	Block
SMTP missing	Email test/schedule fail
WhatsApp token missing	Test fail
Camera permission denied	Scanner error (mobile)
2FA phone kho gaya	Recovery code / admin reset
Password bhool gaya	Admin reset
User inactive	Login + sessions block
Role change	Agli request se naya role
Backend band	Error panel
Port busy (desktop)	Agla free port (5051…)
DB delete	Backup se restore
Multi-source local	File-based merge
Sheet security local	Hidden (GAS only)

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11. Troubleshooting

PROBLEM	SOLUTION
Page nahi khul raha	`http://localhost:5050/api/health` check
API error	Setup/Connection me URL; desktop pe default
Android connect nahi	Wi-Fi same, `adb reverse tcp:5050 tcp:5050`
Login fail	Caps Lock, 10+ policy, admin reset
Account locked	15 min wait
2FA code galat	Phone time auto; recovery code
2FA reset	Admin reset
Email nahi ja raha	SMTP host/from/pass → test
WhatsApp nahi	Token, phone ID, country code
Import fail	Barcode column, size, errors preview
Rollback nahi dikh raha	Committed job hi rollback hota hai
Charts khali	Inventory empty / process nahi chala
Scanner kaam nahi	Terminator setting, min length, focus
Serial scanner nahi	Port/baud check, desktop app use karo
Print bigad raha	Margins None, scale 100%
Nav double highlight	1.0.1 update karo (fix included)
Windows SmartScreen	More info → Run anyway
macOS "unverified"	Right-click → Open

12. Security checklist

- [ ] Owner password strong (10+, letters+numbers)
- [ ] 2FA enable
- [ ] Extra users deactivate
- [ ] Har hafte sessions/audit check
- [ ] Backups safe location
- [ ] Public internet ho to HTTPS reverse proxy
- [ ] SMTP/LLM keys rotate
- [ ] Employee exit pe sessions revoke

13. Useful commands

```
# server dev
cd server && npm run dev

# build
cd server && npm run build

# tests
cd server && npm test

# panel rebuild
cd server && npm run build:panel

# admin user
cd server && npm run create-user -- owner@example.com 'StrongPass123' ADMIN

# Docker
docker compose up --build

# Android APK
cd mobile/android && ./gradlew assembleDebug

# Desktop packages
cd desktop && npm run dist:mac
cd desktop && npm run dist:win:x64
cd desktop && npm run dist:win:arm64
```

14. FAQ aur glossary

FAQ

Data kahan? Local: `bulkrepair.db`. GAS: Google Sheet.

Internet? Local me nahi; GAS me haan.

Multi-warehouse? Abhi single global inventory.

Hindi/Gujarati? Sidebar bottom selector; default English.

Data delete? Backup restore / import rollback.

Kitne devices? Unlimited; sessions page se revoke.

Scanner driver? USB HID ko nahi chahiye; serial ke liye OS driver.

Glossary

TERM	MATLAB
Barcode	Device ka unique ID
Bucket	Category (age/status/location)
Priority	P1/P2/P3 urgency
Source	Data ka origin
Mapping	Column → field
Staging	Import ka temporary area
Job	Ek import/process run
Conflict	Do sources ki alag value
Rollback	Committed import undo
Bin bucket	Shelf/cage group
Session	Login token
2FA	Authenticator code

15. 1.0.1 me kya naya hai

AREA	CHANGE
Sidebar	Legacy ke saare **28 tabs**
Naye pages	Ageing Trends, Processing, Exports, Email Settings, Security, System Health, Help
Focused pages	Report Templates, Report Builder, Multisource, Activity Log, Processing History, Data Quality
Nav fix	Parent/child route double highlight fix
Build	Har platform ke liye panel copy guaranteed
Scanners	Hardware HID + serial support (desktop)
UI	Full light theme, #0078FF, 4px gutters, 6px cards, thin typography
Forms	Har field me example/placeholder
Docs	Ye guide + PDF

End of guide. Koi cheez samajh nahi aayi ho ya screen match na kare to app ka screenshot bhejo — main us hisse ko update kar dunga.